# Fahad Md Kamal

**Senior Software Engineer** | Python, Go, TypeScript | Distributed Systems & Cloud | AI/LLM

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## SUMMARY

Senior software engineer with 6+ years shipping production systems across fintech, healthcare, cybersecurity, geospatial, customer engagement, and enterprise SaaS, with deepest strength in backend and distributed systems. Builds Python (Django, FastAPI) and Go microservices on AWS, Terraform, and Docker with strong security foundations (RBAC/IAM, MFA, encryption, audit logging), and takes features end to end with React/TypeScript front ends. Delivered ~30% faster API responses, pipelines processing 5M+ records, and 99.9% availability for 3,000+ concurrent daily users. Ships LLM features to production (a RAG workflow that cut manual reporting by 60%) and designs human-in-the-loop AI.

## CORE SKILLS

**Languages:** Python, Go, SQL, JavaScript / TypeScript

**Backend & APIs:** FastAPI, Django, Django REST Framework, Gin, REST & gRPC API Design, WebSocket APIs, Microservices, Auth (JWT, OAuth2, TOTP 2FA)

**Databases & Data Modeling:** PostgreSQL (schema design, query optimization), Redis, PostGIS, MySQL, Elasticsearch, Cosmos DB, Normalized & Multi-Tenant Data Models

**Cloud & DevOps:** AWS (ECS, EC2, Lambda, RDS, S3, SQS, Athena, KMS), Terraform, Docker / Compose / Swarm, CI/CD Pipelines, Nginx, Azure (OpenAI, Speech-to-Text, Cosmos DB)

**Distributed Systems:** Event-Driven Architecture, Message Queues (AWS SQS, Redis Pub/Sub, Celery), Idempotency, Concurrency, Serverless, Multi-Tenant SaaS

**Domain Experience:** Fintech & Payments (Stripe, HMAC-Verified Idempotent Webhooks, Ledgers, Invoicing, Reconciliation), Healthcare (Consent-Gated Records, Clinical AI), Cybersecurity Risk, Geospatial Data, Business Intelligence, Customer Engagement, Public Health

**Security & Compliance:** RBAC / IAM Design, MFA, Envelope Encryption (AWS KMS), Audit Logging, Consent-Gated Access, Privacy-by-Design for Sensitive Data (PII/PHI)

**Reliability & Testing:** Health / Readiness / Metrics Endpoints, Prometheus / Jaeger, Structured JSON Logging with Request Correlation, pytest / pytest-django, Code & Security Reviews, Git/GitHub, Agile/Scrum

**AI & ML:** LangChain, RAG, Azure OpenAI, OpenAI / Claude APIs, LLM Tool Use & Structured Extraction, Human-in-the-Loop AI Design, OpenCV, AI-Assisted Development

**Frontend & Mobile:** React, TypeScript, Vue.js, Flutter

## PROFESSIONAL EXPERIENCE

**Senior Software Engineer — BJIT Limited** *Jul 2023 – Present*

*Dhaka, Bangladesh*

- Engineered scalable DRF and FastAPI microservices across concurrent client engagements in cybersecurity, geospatial, business intelligence, and AI document processing, improving API response times by ~30% through Elasticsearch-backed query optimization and performance tuning.
- Turned business and AI workflow requirements from product, front-end, AI, and client teams into performant, well-tested APIs in Agile/Scrum, and raised team standards through code reviews.
- Provisioned AWS infrastructure with Terraform and shipped containerized services through CI/CD to Docker Swarm on EC2, plus serverless AWS Lambda/SQS pipelines.

**Senior Software Engineer — Mevrik DCX** *Dec 2021 – Jul 2023*

- Designed and delivered the REST and real-time APIs behind the user-facing features of a multi-tenant engagement platform serving 3,000+ concurrent daily users.

**Software Engineer — Alesha Tech Ltd.** *Feb 2021 – Nov 2021*

- Owned delivery, from requirements through deployment, of in-house ERP and B2B REST APIs in Python/Django and backend services for Alesha CRM and Alesha DCX (digital business cards).

**Intern Software Engineer — Dream Array** *Sep 2020 – Dec 2020*

- Built Django REST APIs and a normalized PostgreSQL schema for a maternal health platform (appointments, reports, health tracking), deployed on AWS EC2/RDS with CI/CD.

## KEY PROJECTS

**Ledgerly — Full-Stack SaaS Accounting Platform — Solo Developer** *Mar 2025 – Present*

- Built a full-stack, multi-tenant SaaS accounting platform end-to-end: Go (Gin/GORM) and PostgreSQL backend with a React/TypeScript front end: chart of accounts, journal entries, invoicing, bank reconciliation, and AR/AP aging.
- Integrated Stripe Checkout and a billing portal with HMAC-verified, replay-protected (idempotent) webhook handling, keeping payment and ledger data correct under retried and duplicate events.
- Designed a deny-first IAM/RBAC system with a visual "IAM Studio" and audit logging on every policy change, backed by Redis caching and background jobs.
- Added production observability: /healthz, /readyz, and /metricsz endpoints and request-correlated JSON logs.

**Cyber Risk Rating Platform — BJIT Limited** *Jul 2023 – Aug 2024*

- Built backend microservices processing 5M+ records, with Elasticsearch-powered threat-intelligence search and event-driven real-time updates over Redis Pub/Sub.
- Secured access with JWT + TOTP two-factor authentication and built OAuth2 partner integrations.
- Split the platform into a Django/DRF core (API, SQS worker, PDF renderer) and FastAPI gateway and scoring microservices on Docker Swarm/EC2, provisioned with Terraform.

**Enterprise Geospatial Data Platform — BJIT Limited** *Jul 2025 – Sep 2026*

- Designed the Django/PostGIS backend and an event-driven, queue-based serverless ingestion pipeline (AWS Lambda/SQS/Athena) that converts GIS uploads to GeoParquet for large-scale querying.
- Led a security review that resolved injection and token-handling risks; implemented RBAC with group-based file sharing, a callback-authenticated Lambda-to-Django handoff, and PostGIS databases isolated from the primary business database.

**AI Document Processing Tool (RAG) — BJIT Limited** *Mar 2025 – Jun 2025*

- Built and shipped to production a LangChain and Azure OpenAI RAG workflow that turns audio/video recordings and PDFs into structured business reports (Azure Speech-to-Text, Cosmos DB), cutting manual report generation by 60%.

**AEGIS — Healthcare Identity & Patient Records Platform — Solo Developer** *Jan 2026 – Present*

- Building a multi-tenant healthcare identity and patient records platform in Go (gRPC), PostgreSQL, Redis, and Python/FastAPI on AWS/Terraform, covering identity and access management, patient records, provider access, and visit workflows.
- Designed for privacy from the start: encryption, audit logging, patient-controlled (consent-gated) access, RBAC, and MFA.
- Building AI-assisted visit documentation (speech transcription and Claude) that drafts clinical notes for clinician review, with a human-in-the-loop gate: no AI output is saved without clinician approval.

**Geographic Business Intelligence Platform — BJIT Limited** *Sep 2024 – Feb 2025*

- Built map-based BI backend services with parallelized multi-database queries and heat-map generation, reconciling two industry-code vocabularies across reference-data and ledger schemas.

- **Omnichannel Customer Engagement Platform — Mevrik DCX** *Dec 2021 – Jul 2023*

- Built backend services for an omnichannel engagement platform (email, social, live chat, chatbot) serving 3,000+ concurrent daily users at 99.9% availability, with JWT-based multi-tenant auth and Celery/Redis async jobs.

- **B-DOPS — Dengue Outbreak Prevention Surveillance — Lead Developer** *Sep 2019 – Jan 2020*
- Led end-to-end development of a mobile-first public health surveillance platform (Django REST + Channels real-time backend, Flutter client) with OpenCV-based triage of citizen-submitted photos.

- **EDUCATION**

BSc (Hons) Computing, First Class Honours — University of Greenwich, UK (NCC Education pathway via Daffodil International Academy, Bangladesh) | Jan 2017 – Feb 2021